Non-Chemical (Water Treatment) Devices

Chemical Replacement Technologies

Technology Discussion, Comparison and Business Approach

April 2022

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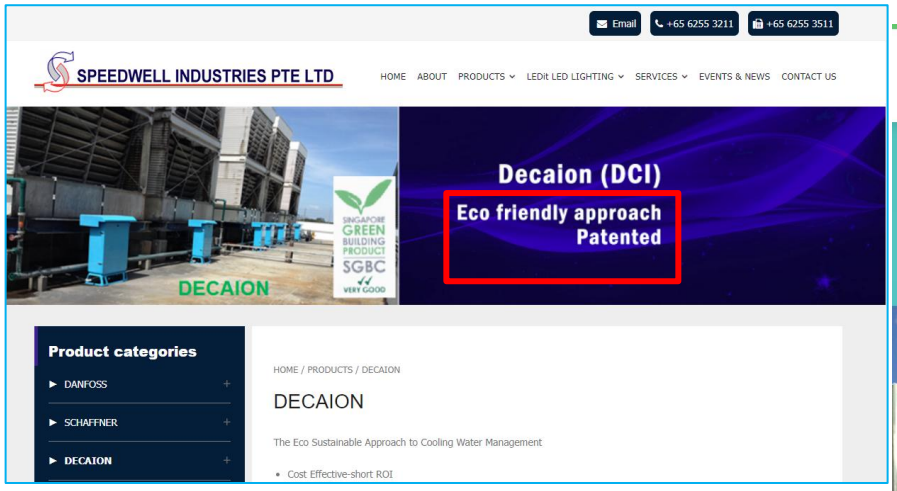
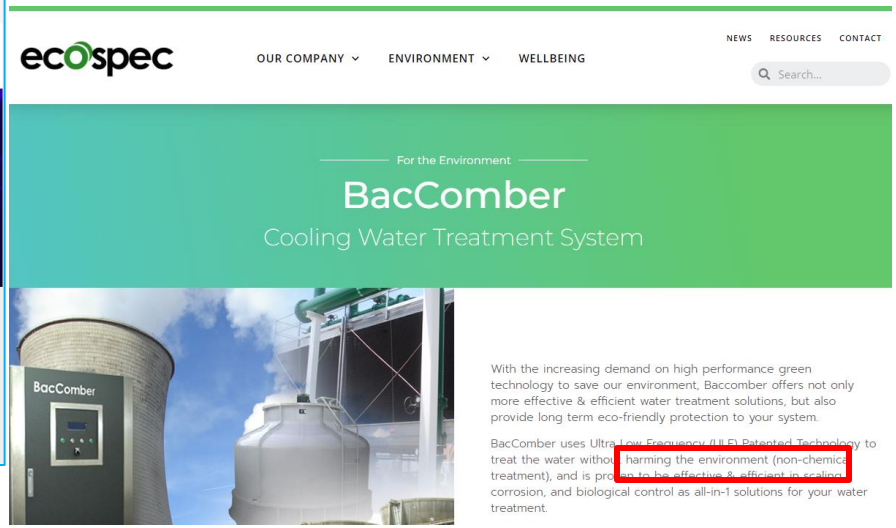
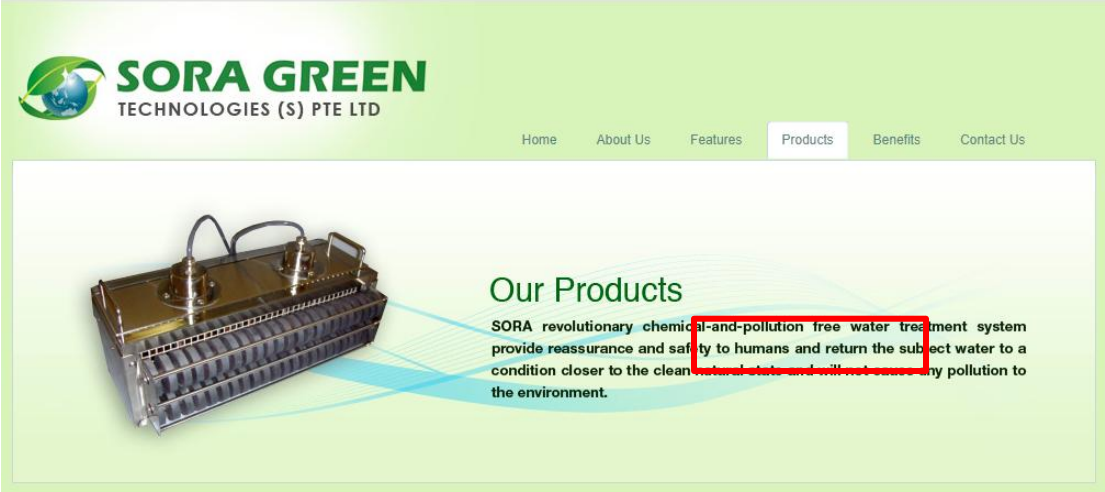
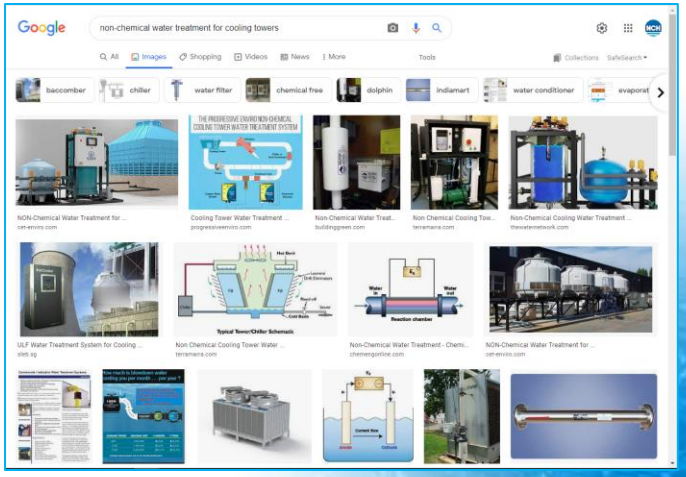
Non-Chemical Devices









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Competition in Sales Process

Design	Engineering	Operation	Contract Renewal
Consultant Design company	Main contractor Engineering	Main contractor End user/client	Main contractor End user/client
NCD in design Client request NCD optional	In the design Recommended NCD optional	NCD in line NCD optional Competition from NCD	More green More saving NCD in line NCD optional Competition from NCD
Building/Hotel	University	Manufacturing	Data Center

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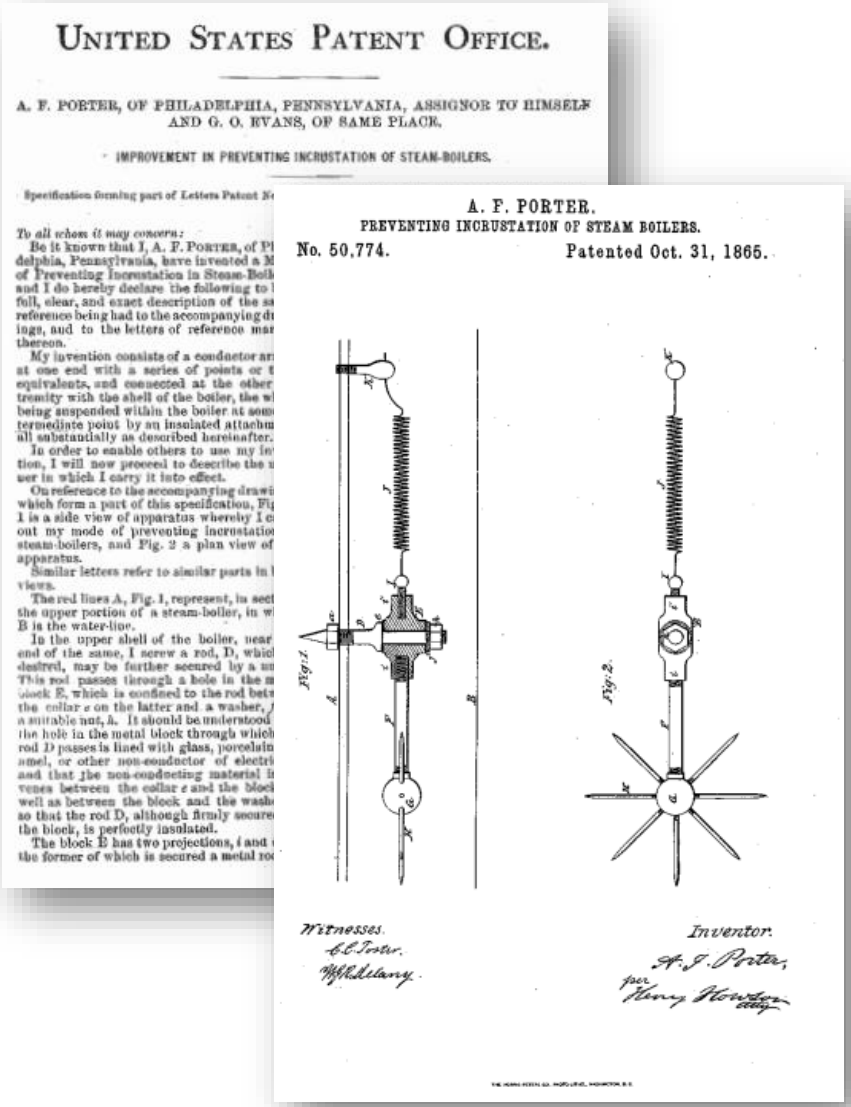
The oldest NCD patent



UNITED STATES PATENT OFFICE.

A. F. PORTER, OF PHILADELPHIA, PENNSYLVANIA, ASSIGNOR TO HIMSELF AND G. O. EVANS, OF SAME PLACE.

IMPROVEMENT IN PREVENTING INCRUSTATION OF STEAM-BOILERS.

A. F. PORTER.
PREVENTING INCRUSTATION OF STEAM BOILERS.
No. 50,774. Patented Oct. 31, 1865.

Fig. 1. Fig. 2.

Witnesses.

Inventor.

- 1865 patent, Improvement In Preventing Incrustation of Steam-Boilers.
- This is the oldest Non-Chemical device for water treatment that we know of.
- Its function is to “collect whatever electricity may be generated within the boiler” and ground it to the shell of the boiler to prevent scale.

https://www.richardhouriganinc.com/resources/non-chemical-devices

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Non-Chemical Devices (NCDs)

- Mechanical water treatment devices used to help control corrosion, deposits, and/or microbiological growth in water
- Some based on well established principles & have predictable results:
  - Deaerators & heated feedwater tanks
  - Centrifugal, screen & media filters
  - Activated carbon filters & adsorption/catalytic filters
  - Reverse osmosis, ultra & nanofiltration
  - Ozone generators & ultraviolet light
  - Cathodic protection
  - Ion exchange processes
  - Electrolytic generation of chlorine from salt
  - Copper/silver ionizers
- Designed to work with chemicals, not a “miracle” replacement

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Non-Chemical Devices

- Also a lot of NCDs on the market that make broad claims that cannot be substantiated
  - Marketing materials rely heavily on testimonials
  - Proposed mechanism of operation generally not consistent with established scientific principles
  - Results difficult to quantify/repeat in controlled studies
  - If these devices were as effective as claimed, major consumers of water treatment would have adopted long ago
- We will examine the various types of NCD technology, try to explain the basic working principles, & discuss the relative merits of each approach or device

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Technology Comparison

A few key points to remember for ALL NCD's:

- The rules of nature and science must apply to every device (Mother Nature does not play favorites)
- For an NCD to work, at least one chemical reaction has to occur
  - Chemical reactions take energy, whether mechanical, electrical, thermal or chemical energy
  - Be very suspicious of any NCD that claims to use no energy or very minimal energy
- No NCD will work in every type of water or system or do all things
- The definition of what "works" & is "acceptable" results is subjective!

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Technology Comparison

When you need to apply a complete solution:

- None of the NCD's currently on the market provide any treatment solution for closed systems or boilers
  - These systems must be chemically treated
- No NCD is adequate for pre-commission cleaning/passivation or system lay-up
  - These needs must be met with traditional chemicals
  - Most NCD distributors generally lack this capability
- Chemical free does not mean service free
  - All of the NCD's that actually work are complex devices that require regular service from a trained, competent representative (Deaerators, filters, UV light…)

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Non-Chemical Devices

Let's start by looking at the general types of technologies being used by NCD's. You can divide all these technologies into 6 general categories:

- Magnetic, Electromagnetic, & Electrostatic Devices
- Pulse Power Devices
- Electrolytic Ion Generators
- Sidestream Precipitation (not really non-chemical)
- Cavitation (Ultrasonic & Hydrodynamic)
- Catalytic Devices

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Magnets, Electromagnets, & Electrostatic Devices: The Claims

- Have been around since the late 1800's
- **Magnets** - Employ one or more permanent magnets mounted on the outside or inside of the pipe containing the water to be treated

  (磁铁, 磁石, 자석, แม่เหล็ก, Nam châm)
- **Electromagnets** - Low voltage power supplied to coils wound around the outside of the pipe containing the water to be treated to produce a magnetic field

  (电磁铁, 電磁石, 전자석, แม่เหล็กไฟฟ้า, Nam châm điện)
- **Electrostatic devices** – High voltage, low current applied to rods or plates in cooling water to create an electrostatic charge or field in the water being treated

  (静电, 静電, 정전기, ไฟฟ้าสถิต, Tĩnh điện)

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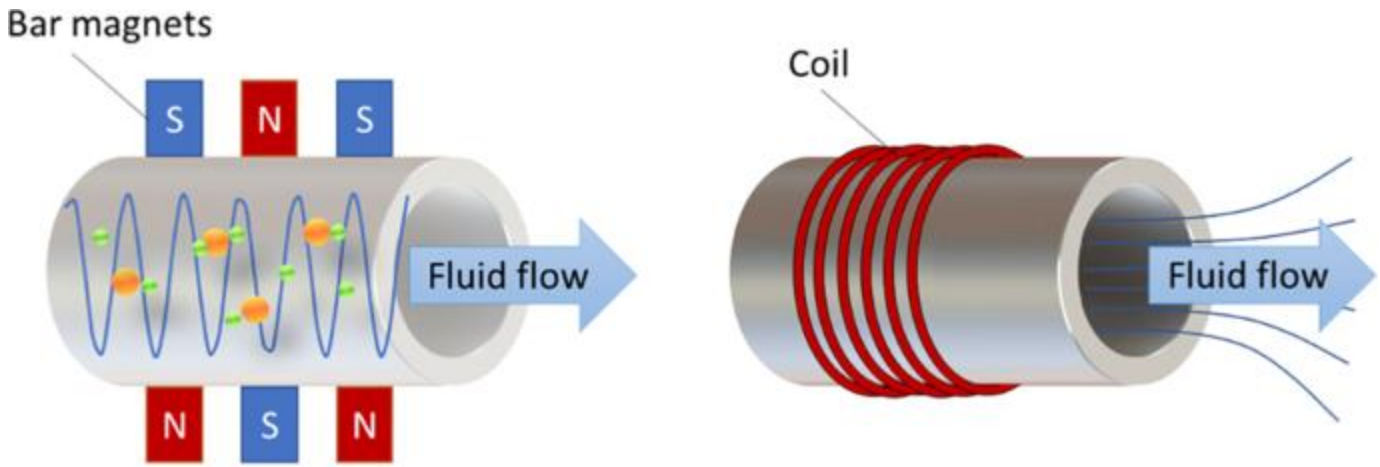
Magnets, Electromagnets, & Electrostatic Devices: The Claims



(a) Permanent magnets

(b) Solenoid coil

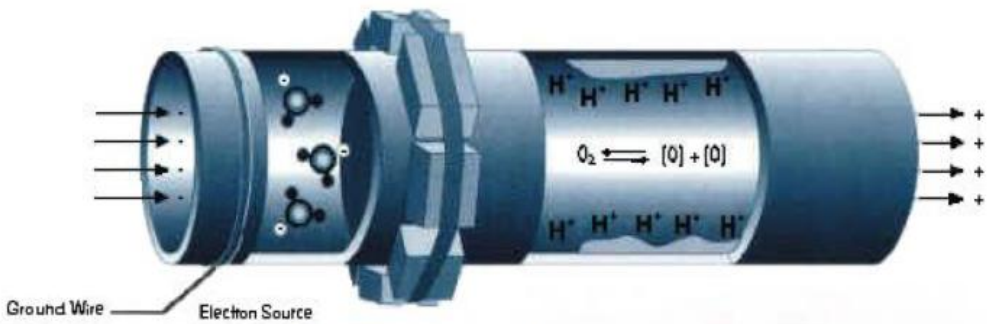


Illustration of claimed magnetic device operation

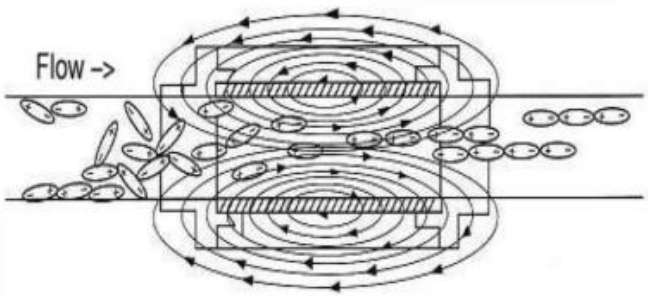


Illustration of claimed electromagnetic device operation

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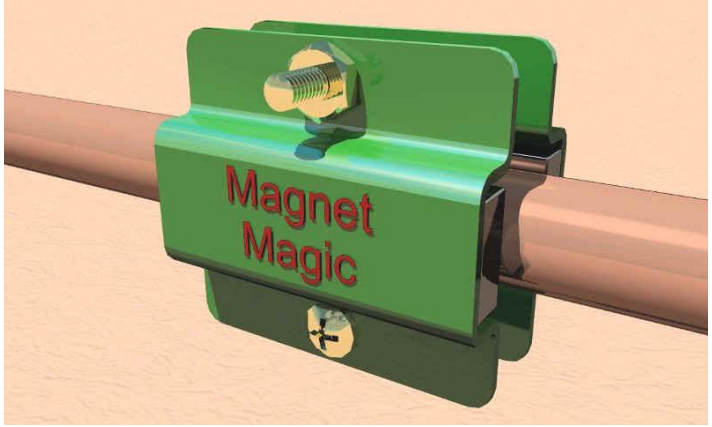
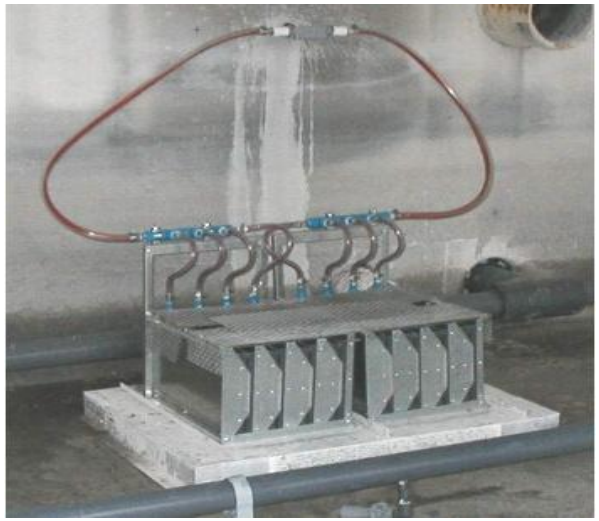
Magnets, Electromagnets, & Electrostatic Devices



Generic

Scale Blaster



Fluidtron

Ion Stick

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Magnets, Electromagnets, & Electrostatic Devices: The Claims

- Claim that by passing water through a magnetic or electrostatic field, they can change the ionic charge on the scale forming ions in the water
- Often sold as "plug and play" device, no adjustment once installed
- Some claim to change the ionic charges so that they are all the same, causing the calcium & carbonate ions to repel each other so that scale does not form
- Some claim to change the ionic charges so that they "align", causing the calcium & carbonate ions to attract each other so that precipitation forms in the bulk water
- Sounds pretty good doesn't it?

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Magnets, Electromagnets, & Electrostatic Devices: The Real Science

- A flowing water stream is a grounded environment; any magnetic or electrostatic charge placed on particles or molecules would immediately short to ground
- But, even if you could….
  - It would take huge amounts of energy to make this happen!
  - If you have 100 ppm of calcium in your tower water, there would be 3.6×10^{20} ions of calcium in ONE OUNCE (0.03L) of your tower water
  - That's 3,600,000,000,000,000,000,000 ions
  - It's impossible for any low energy magnetic or electrostatic device to change the charge on a ions in water
- There is nothing about any low power magnetic or electrostatic field that would have any effect on bacteria or algae growth
- There is nothing that is claimed about changing ionic charges that would offer any demonstrable protection against corrosion

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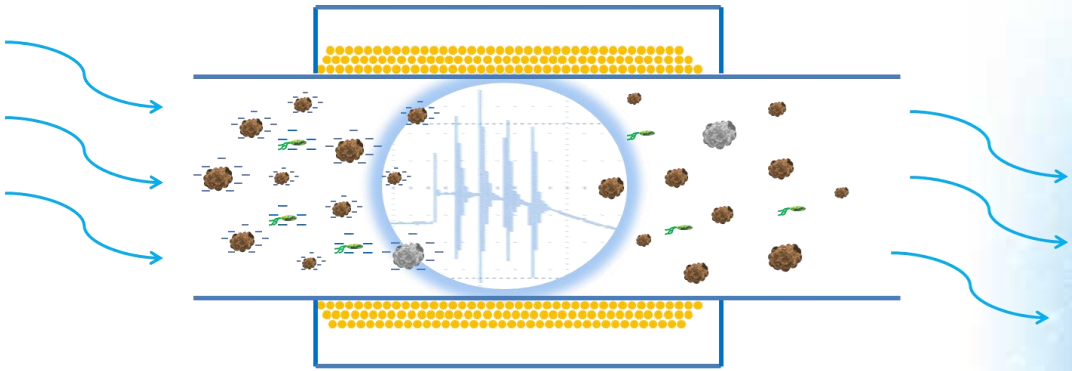
AC Induction/Pulse Power: The Claims

- These units use a "pulse power" coil, which is basically an electromagnet subjected to a variable voltage power source
- They claim that the pulsing electromagnetic field activates "colloidal nucleation sites" causing bulk water precipitation of calcium carbonate which settles out in the tower basin
- They claim that the way in which the calcium carbonate scale forms traps any bacteria by encapsulation & removes them from the bulk water
- 3" Dolphin rated for 1000 gpm cost Mechanical Contractor $15,000

(脉冲电场, パルス電界, 펄스 전기장, สนามไฟฟ้าพัลส์, Điện trường xung)



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AC Induction/Pulse Power Units

Dolphin

Griswold

Pulse Pure

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AC Induction/Pulse Power: The Real Science

- Low power magnetic fields do not impart enough energy to the system to effect any ionic changes
- Planktonic bacteria are free swimming and can not be encapsulated (If this were true, then a heavily scaled system would have no bacteria. In the real world you find the exact opposite)
- No mechanism for control of sessile or anaerobic bacteria (biofilm)
- No mechanism for corrosion control

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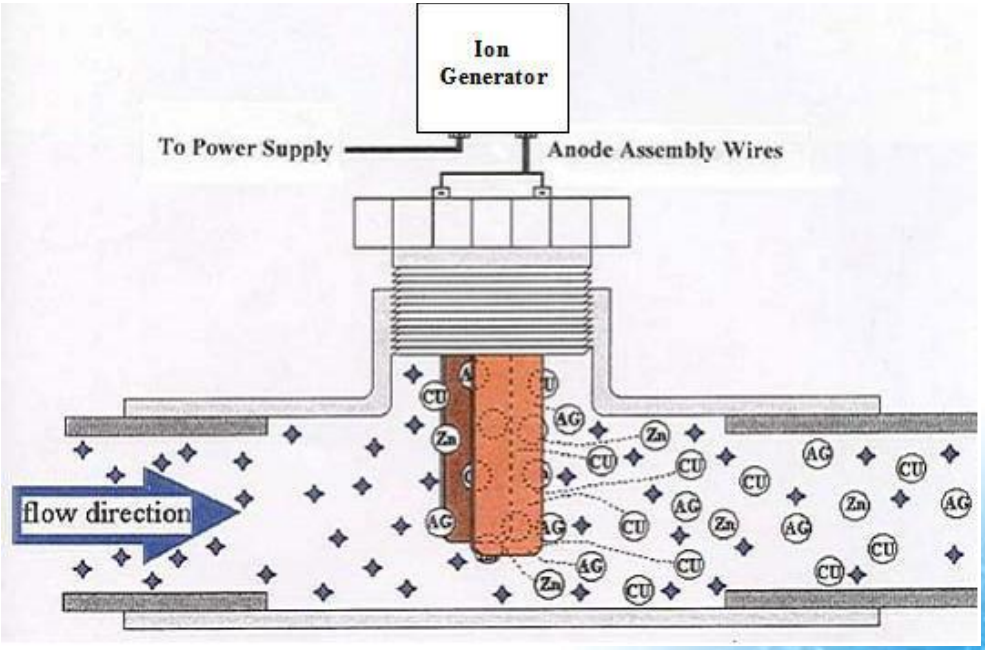
Electrolytic Ion Generators: The Claims

- The system use the scientifically valid process of electrolysis, where a current is run through an electrode in the water. The current causes ions to be released from the electrode into the water
- The most common types of electrolytic ion generators include:
  - Copper/Silver (Cu/Ag)
  - Zinc
- Claim copper & silver ions are effective biocides
- Claim zinc ions are effective for corrosion control

(电解离子发生器, 電解イオン発生器, 전해 이온 발생기, เครื่องกำเนิดไอออนด้วยไฟฟ้า, Máy tạo ion điện phân



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Electrolytic Ion Generators: The Real Science

- From a scientific standpoint, these units actually do some of what they claim
- Copper/Silver systems have proven success as a biocide
  - Good in swimming pools and hot tubs and some applications in secondary disinfection of potable (drinking) water
  - Not very successful in large systems with heavy MB loading such as a cooling tower
- However:
  - Copper and Silver are heavy metals, and toxic in the environment*
  - Nothing for scale control
  - Nothing for steel/copper corrosion
  - Copper can plate on steel/galvanized & cause galvanic corrosion

*** Similar to mercury poisoning**

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Electrolytic Ion Generators: The Real Science

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Electrolytic Ion Generators: The Real Science

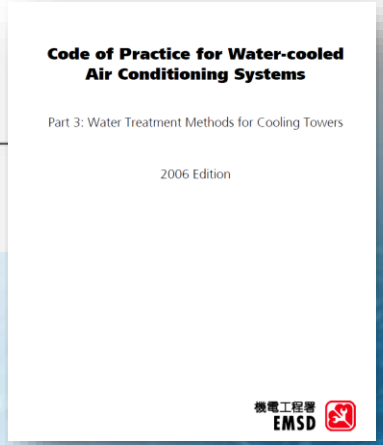
CHEM-AQUA

4.3.2 Copper and Silver Ionisation

4.3.2.1 Ionisation indicates the electrolytic generation of copper and silver ions in cooling tower water. If properly managed, copper and silver ion concentrations at 20 to 30 µg/L and 10 to 15 µg/L, respectively, can be effective to kill bacteria in the systems. The ions assist in the control of bacterial populations in the presence of a free chlorine residual of at least 0.2 mg/kg.

4.3.2.2 It should be noted that in hard water systems, silver ion concentrations is difficult to maintain due to build-up of scale on the electrodes, and the high concentration of dissolved solids precipitating the silver ions out of solution. For both hard and soft water, the ionisation process is pH sensitive and it is difficult to maintain silver ion concentrations above pH 7.6. It is not recommended to adopt ionization in systems having steel or aluminium heat exchanger since deposition of the copper ion and

2006 Edition (Jan 2007) 17 EMSD



機電工程署 EMSD

Electrical and Mechanical Services Department
The Government of the Hong Kong Special Administrative Region

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Electrolytic Ion Generators: The Real Science

- Electrolytic Zinc Generator
  - Zinc can be effective corrosion inhibitor for mild steel
  - Zinc sacrificial anodes are commonly used for pipeline protection in the oil & gas business
- Not chemical free
- Can present discharge problems
- Zinc provides no corrosion protection for copper, brass or bronze
- Zinc has no effect on bacteria or algae

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Combination Devices

- Use a combination of NCD and chemical technologies in a “package”
- Envirotower
  - Magnetic/Electrostatic Conditioner
  - Mineralator (releases Iodine & Zinc)
  - Centrifugal separator (small)
- NEXTEQ
  - Magnetic/Electrostatic Conditioner
  - Copper/silver ionizer
  - Zinc electrode

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Side Stream Precipitation: The Claims

- There are two general classes of side stream precipitation devices:
  - Chemical
  - Electric anode
- Chemical units claim to use a side stream softening process to remove some calcium & most of the silica from the water. They also claim to remove some of the alkalinity
- Electric anode units claim to use electrolysis to precipitate calcium at an anode. Since this precipitates as calcium carbonate, some alkalinity reduction also occurs
- Both types of these devices are capable of doing what they claim

(electrolysis, 电解, 電解, 전기분해, อิเล็กโทรลิซิส, điện phân)

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Side Stream Precipitation: The Real Science

- Continuous stream precipitation softening has been in use since before WW II. It was developed for boiler water pretreatment & was widely used until it was replaced by sodium zeolite softening, which was more effective.
- Continuous side stream precipitation softening removes most of the calcium, most of the silica & lowers the alkalinity
- Does nothing for corrosion control
- Does nothing for microbiological control
- Creates a continuous waste stream of precipitated calcium sludge
- Chlorides can build up to high levels
- Not really non-chemical

Chemico HiCycler

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Side Stream Precipitation: The Real Science

- Electrochemical precipitation at an anode or cathode is a well documented scientific process
- Removes only calcium, does not remove silica
- Some systems that precipitate calcium carbonate also reduce the total alkalinity
- Does nothing for corrosion control
- Does nothing for microbiological control
- Electrodes need to be cleaned or replaced at regular intervals

HOH Green Machine

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Cavitation Devices: The Claims

- There are currently two general types of cavitation devices. These are:
  - Ultrasonically generated cavitation
  - Mechanically or hydraulically induced cavitation
- Using either ultrasound waves or fluid mechanics, cavitation is induced in a side stream of water
- Cavitation is formation and collapse of small bubbles in a fluid stream
- When the bubbles collapse, strong pressure waves are created & localized high temperatures are generated

  (Cavitation, 涡流效应， キャビテーション， 캐비테이션， คาวิเทชั่น， Xâm thực）

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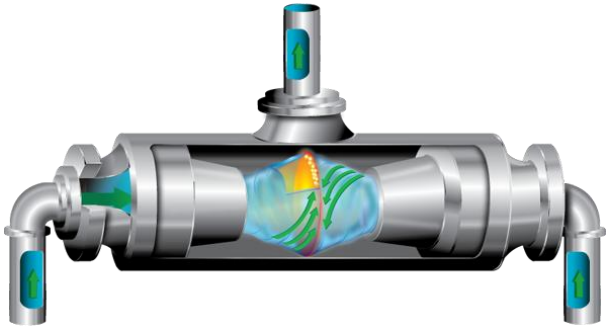
Cavitation Devices: The Claims

- Ultrasonic & hydrodynamic cavitation systems
  - When the pressure waves occur, stresses or destroys the bacteria cell wall. Thus killing the bacteria by cell lysis.
- Hydrodynamic cavitation systems
  - Pressure wave & temperature spikes cause the bicarbonate alkalinity to convert to carbonate plus carbon dioxide gas
  - Causes flash precipitation of calcium carbonate crystals which can be removed by filtration
  - The CO_2 gas flashes off & buffers the pH of the system to 8.5 to 9.0 resulting in “carbonate cycle” corrosion protection



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Cavitation Devices: The Real Science

- Cavitation really occurs with these units
  - Cavitation chemistry is a widely accepted scientific principle with many additional applications outside of water treatment
  - The ability of cavitation to kill planktonic bacteria in water has been extensively proven and is widely accepted
  - The flash precipitation of calcium carbonate has been proven in many laboratory studies and demonstrated in real world situations
  - The amount of energy generated by the devices impacts the amount of cavitation & their overall performance
- The benefits of filtration are well documented

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Cavitation Devices: The Real Science

- There is nothing in the cavitation based systems for corrosion control of copper
- Mild steel corrosion control generally not be as good as with a high performance chemical inhibitor program
- Galvanized steel must be passivated to withstand high pH operation
- There is nothing in the cavitation based systems for elimination of sessile bacteria (biofilm)
- There is nothing in the cavitation based systems for control of algae

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Cavitation Devices

VRTX

Sonoxide

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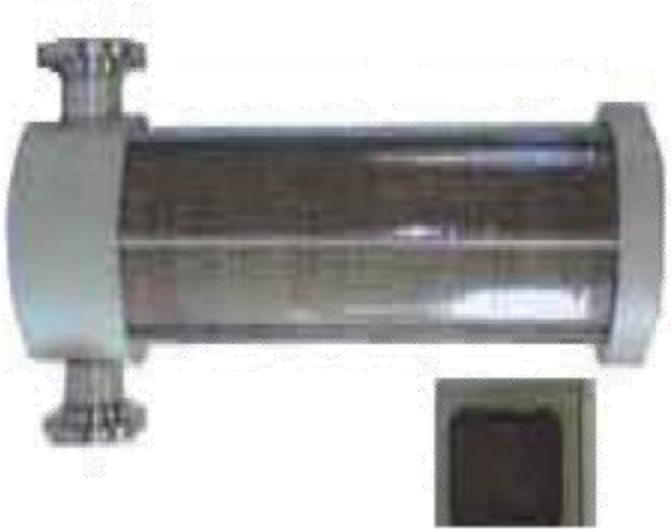
Catalytic Devices

- These devices appear to be nothing more than sealed metal cylinders filled with "powdered metal", or metal shapes of varied composition,
- inserted into either a tank, or a pipe, in the water system to be treated.
- The chemical composition of the "catalyst" is reported to vary widely, but is often claimed to include transition and precious metals.

（催化， 触媒作用， 촉매， ตัวเร่งปฏิกิริยา， Xúc tác）



Typical catalytic device

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How the distributors of NCD's claim that they work and why it's false

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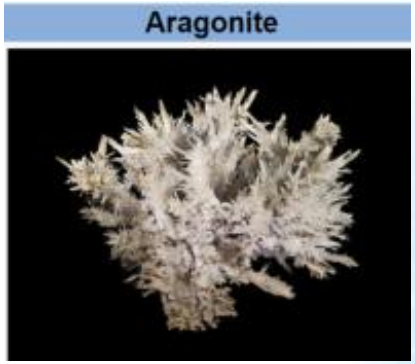
Testimonials vs Science

- Most NCD distributors have one thing in common. They claim that their device forces the formation of **Aragonite** instead of **calcite** and Aragonite will precipitate in the bulk water and not form scale.
- This sounds scientific but is not true.
- Aragonite can't form in cooling tower water at that temperature and pressure and even if it did, Aragonite would make a scale that is even denser then calcite so it would not be desirable.
- In fact, aragonite is the low temperature stable phase of calcium carbonate with a reported transition temperature of 128° C (262.4° F) to calcite. Some reversion is reported in the literature, at a temperature of -60° C (-76° F).



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Testimonials vs Science

- Other NCD sellers claim their device **alters the charge** on the particles or molecules. But cooling water is very conductive and any charge put on a particle or molecule would immediately short to ground.
- Also, calcium and carbonate exist in water as dissolved ions and not in particle form. Exposure of these ions to magnetic, mechanical or electrical forces will not alter their ionic state.
- They may also claim that the device changes polarity alignment. Even if this happened it would immediately be destroyed by simple Brownian motion.

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Testimonials vs Science

- Other claims are removal of carbon dioxide and oxygen from the water reducing scale forming tendencies. This is in fact false. Removing these does not affect the crystal form that calcium carbonate will take in precipitation or the LSI?
- High shear in mechanical devices claim to crush organisms. They may crush planktonic but the real problem in cooling systems in sessile bacteria and not Planktonic.
- Most claim Bulk precipitation in tower basin water with high cycles causing high pH. This can be forced by turning off the blowdown but causes scale and sludge.

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Scientific Evaluations of NCD's

- AIR FORCE CIVIL ENGINEERING SUPPORT AGENCY

  "Electronic or magnetic devices for scale control have not been proven effective and should not be used."

- FACILITIES ENGINEERING, U.S. Army Center for Public Works

  "Therefore, the use of these nonchemical treatment devices on Army systems either regularly or on a test evaluation basis, is prohibited."

- **3M** results of 10 year study of NCD's

  "These devices showed no significant effects in reducing the scale forming in side by side steam heated tube shells."

- **ASHRAE** The American Society of Heating, Refrigerating and Air Conditioning Engineers

  "The results of this study conducted under well-controlled conditions show that NCD'S did not control biological growth under the conditions of the testing in the pilot scale cooling tower."

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Business Approach And Consideration

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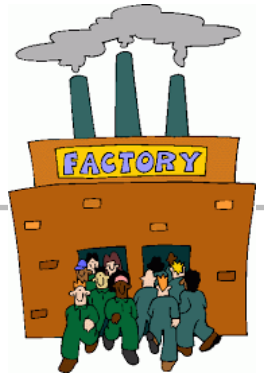
Competition in Sales Process

Design	Engineering	Operation	Contract Renewal
Consultant Design company	Main contractor Engineering	Main contractor End user/client	Main contractor End user/client
NCD in design Client request NCD optional	In the design Recommended NCD optional	NCD in line NCD optional Competition from NCD	More green More saving NCD in line NCD optional Competition from NCD
			
Building/Hotel	University	Manufacturing	Data Center

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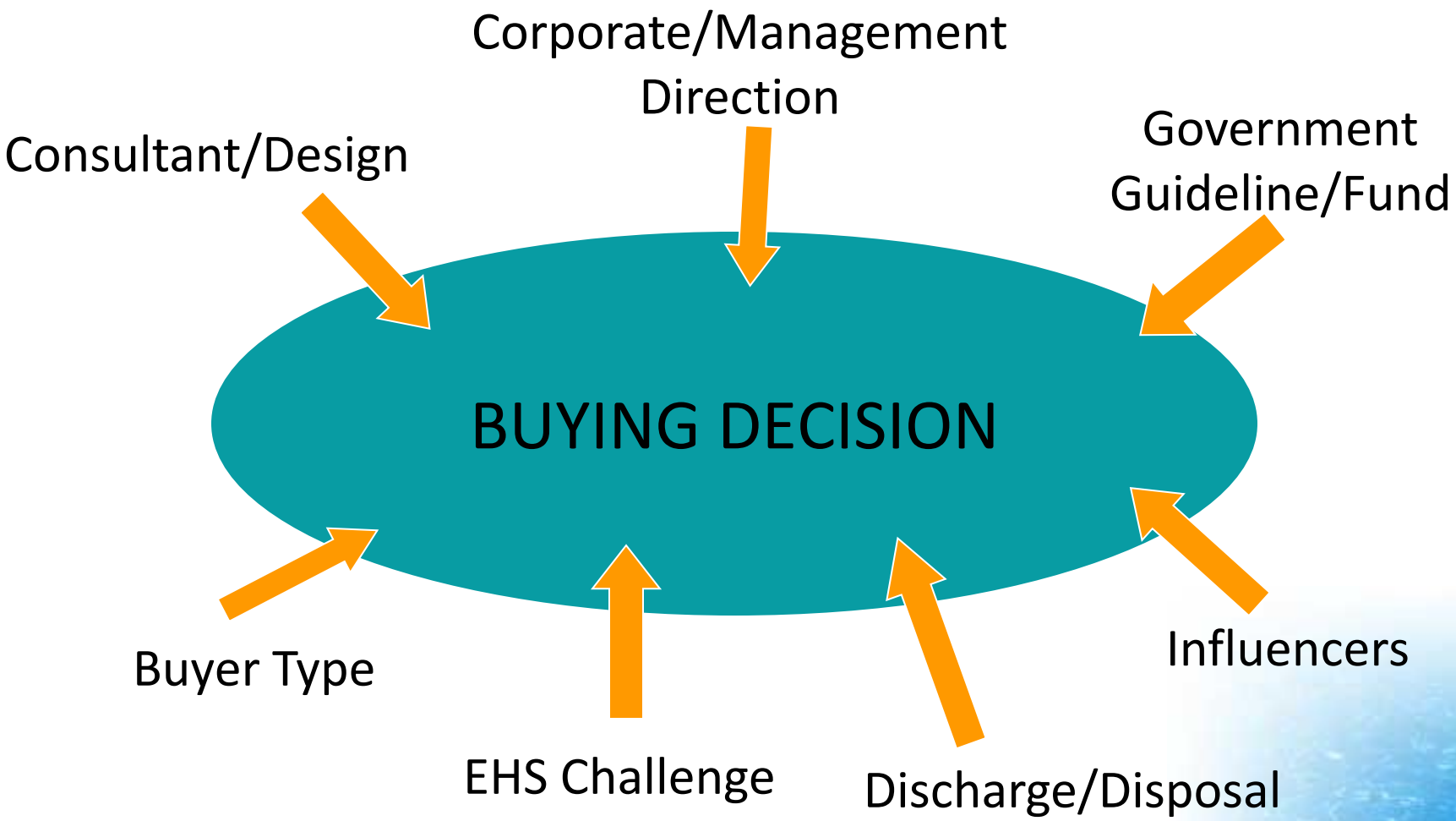
Business Decision Making



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Business Approach

- To be more GREEN, SUSTAINABILITY, ENVIRONMENTAL FRIENDLY (less chemical) is part of the vision of many prospects
- Build up the relationship and trust from the beginning is the foundation
- Clear with the decision making process, decision maker and cycle of influence
- Focus on QB's needs, problem solving and ROI
- Better technical preparation, more patience, different strategy
- To avoid working with NCD together in the same system
- When the NCD attack our business, we need to have a good tactics
  - Defense: client review/trust/communication
  - Counterattack: share the facts with QB

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Quote from paper by Helmut H. Brugman

- "There are a series of basic questions that can be asked of any (NCD) Vendor." But there is one question a possible user must ask himself: **How much risk am I willing to take?**
- The following questions were developed by NACE International.
  1. How do you know that the device is performing the function for which it was installed?
  2. What means will be used to size the equipment or process?
  3. Under what conditions will the equipment or process perform the desired function as intended?
  4. Has an independent testing laboratory such as National Sanitation Foundation or the German DVGW tested and approved the equipment or process for its intended use using a technically sound protocol?
  5. Are there documented case histories of successful applications similar to the intended usage?
  6. What are the costs of installation, maintenance, and operation of the equipment or in situ process and its components?
  7. What is the expected mean time of failure for the product?
- A question every possible user must ask and get the answer in writing: **Does the vendor have sufficient product liability insurance to cover all the damages from the failure?**

The Opportunity

Why not offer a better Greener overall alternative from Chem-Aqua that actually works?

- Reduced water consumption (Higher better controlled cycles), Controllers and PTSA traced, water meters
- More environmentally friendly chemicals, shorter half-life biocides, less Toxic inhibitors and biocides?
- Solid Chemical, reduce environmental impact, Caustic transport, CO_2 reduction, etc.
- Water recycle
- Zero Blowdown
- Other

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Summary

- NCDs have been marketed for control of scale, corrosion and MB growth in cooling water for over 60 years.
- The vast majority of NCD marking is based on the testimonials, rarely is any scientific data provided.
- The most common causes behind successful testimonials are other operation conditions (water quality, CoC, pH, LSI…)
- Understand the science behind the NCD is important for use to defense and attack the challenge and opportunity
- GOS is our basic tool to approach or defense the business
- Understand customer needs, find out the value, offer our green and sustainability solutions

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Reference

RESEARCH REVIEW

ASHRAE Research Project Number 1361-RP
Biological Control in Cooling Water Systems
Using Non-Chemical Treatment Devices

Principal Author: Radisav D. Vidic
Contributing Authors: Scott M. Duda, Janet E. Stout
Project Date: April 1, 2008 – December 31, 2009
Final Report: April 2010 (288 Pages)

Executive Summary

The objective of this investigation was to provide a controlled, independent, and scientific evaluation of several classes of non-chemical treatment devices (NCDs) for controlling biological activity in a model cooling tower system. Five NCDs were evaluated for efficacy in reducing planktonic (bulk water) and sessile (biofilm) microbial populations. The devices included magnetic, pulsed electric field, electrostatic, ultrasonic, and hydrodynamic cavitation.

Two model towers were designed and operated to simulate field conditions. One tower served as the untreated control while the NCD was installed on the second tower. Each device trial was conducted over a 4-week period. Plate counts were used to monitor biological growth. Make-up water for each system was dechlorinated city tap water.

Under the experimental conditions used in this study, no statistically significant difference in planktonic and sessile microbial concentrations was observed between the control tower and a tower treated by any of the five NCDs evaluated in this study. Standard chemical treatment of pilot-scale cooling towers by the addition of free chlorine was able to achieve appreciable reduction in both planktonic (2-3 orders of magnitude) and sessile microbial growth (3-4 orders of magnitude) in these towers. These positive controls were repeated three times throughout the study and the results clearly showed that free chlorine was able to control biological growth in every instance, even after heavy microbial colonization of model cooling towers.

The results of this study conducted under well-controlled conditions show that NCDs did not control biological growth under the conditions of the testing in the pilot scale cooling tower systems.

Biological Control in Cooling Water Systems Using Non-Chemical Treatment Devices

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April 2010

Final Technical Report

April 1, 2008 – December 31, 2009

Principal Author:
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Contributing Authors:
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Water Treatment Gadgets

Technical Bulletin 8-001
Miscellaneous

The Problem
Since the late 1800's, devices have constantly appeared on the market claiming to prevent scale, corrosion, and biological fouling in boiler and cooling water systems without the aid of chemical treatment. These devices are called gadgets by water treatment professionals.

Gadgets supposedly operate on magnetic, electromagnetic, electrostatic, sonic, or some other physical principle. Regardless of the claimed principle of operation, they all have one major feature in common: to date, no gadget has been proven effective in unbiased tests conducted by qualified scientists in the laboratories of major universities, independent research institutes, or government facilities. If gadgets were as effective as their promoters claim, they would have found widespread use by major consumers of water treatment chemicals like the U.S. Navy, large oil and chemical companies, etc.

Additional features shared by gadgets include

- Supporting evidence consists of testimonial letters from end users and self-serving research conducted by and for the promoters and manufacturers of the units.
- Without exception, the results of using a gadget cannot be detected by standard water analyses. This is because gadgets do not alter hardness, alkalinity, or any other chemical characteristics of water.
- On close examination, the claimed principle of the gadget's operation will usually be in conflict with basic scientific laws. For example, gadgets operating on electromagnetic or electrostatic principles claim results in direct conflict with the laws of thermodynamics, which require the expenditure of energy to accomplish chemical or physical changes. Even if the effects claimed for these devices could be confirmed, they would apply only for the few seconds that the water passes through the field of the device. There can be no residual effect on the water.

When Gadgets Seem to Work
On occasion, gadgets seem to work. However, careful investigation usually reveals one of the following conditions

- The system did not need treatment in the first place because the water had minimal corrosion and scale-forming tendencies.
- The system was already being treated chemically and the gadget was being superimposed on top of chemical treatment products, which the gadget supplier recommends to "aid efficiency," "increase polarization," etc. In reality, the chemical additives, not the gadget, are controlling scale and corrosion.
- The bleed or blowdown rate was dramatically increased to reduce scale-forming tendencies.
- Almost any change in how a system is operated or in the system chemistry can cause old deposits to break off for a short time. This can give a short-term indication of system cleanup. However, unless the blowdown rate is drastically increased or chemical treatment is resumed, the typical result of stopping treatment is the formation of hard, very difficult-to-remove scale deposits.

Conclusion
The major appeal of gadgets is eliminating the need to purchase and handle water treatment chemicals. They are sold on the basis of improved workplace safety, lower labor costs, reduced environmental concerns, and easier application. In reality, gadgets do nothing to eliminate scale and corrosion problems and can result in unexpected and costly equipment failure.

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Demonstration and Evaluation of Magnetic Descalers

Kent W. Smothers, Charles D. Curtiss, Brian T. Gard, Robert H. Strauss, and Vincent F. Hock

September 2001

Approved for public release; distribution is unlimited.

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Reference

Case Study Miscellaneous

Pulse-Power Device Failure in Large Medical Center

Background

A large Medical Center on the west coast of Florida was retrofitted with new chillers and cooling towers. The architect/design engineer specified a non-chemical device (NCD) for cooling tower treatment so the bleed could be discharged into an onsite decorative pond. This would enable the facility to eliminate sanitary sewer charges while demonstrating a commitment to sustainability.

The new chiller plant consisted of 4,750 tons of refrigeration with stainless steel cooling towers. To meet the design specifications, a pulse-powered NCD with twin centrifugal particle separators was installed. The NCD provider claimed that, with a routine service program, the NCD would provide effective scale, corrosion, and microbiological control equivalent to chemical treatment.

Results

The Medical Center rigorously followed the recommendations provided for the NCD program. However, despite repeated program modifications, serious problems developed

- The chiller condenser tubes had to be brushed and/or acid cleaned quarterly to maintain heat transfer
- After only five years of operation, the tower fill had to be completely replaced due to massive deposits that blocked water flow and reduced cooling efficiency
- Frequent tower cleaning was required due to microbiological growth and deposits in the tower basin
- Serious corrosion problems developed including leaks in tower basins

Excessively high energy consumption, maintenance costs, and labor were incurred by the Medical Center due to overall failure of the NCD. The NCD failed to produce results for scale, corrosion, and microbiological control equal with that of traditional water chemistry.

Corrosion Resulting from Tower Basin Leaks

Cooling Towers

Scale the Size of a Cell Phone

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CS8-004(1M610)

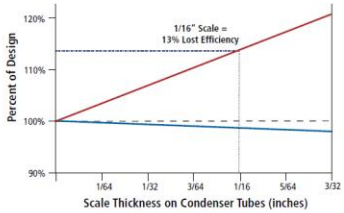
Case Study Miscellaneous

Non-Chemical Device (NCD) Discontinued After Multi-Year Trial

Background

A FORTUNE®* 100 food company installed a non-chemical water treatment device at one of their facilities. The device reportedly makes opposing water streams collide with such force that "hydrodynamic cavitation" occurs. For a cooling tower system, it is claimed this process causes: calcium carbonate to form in the bulk water where it can be removed by filtration, the water to be degassed and made non-corrosive, and microorganisms to be destroyed. Reduced bleed was also promoted.

After a test period, the company decided the NCD would replace chemical treatment in over 20 facilities with the objective of reducing costs and becoming more environmentally responsible.



Condenser Scale Reduces Compressor Capacity and Increases Horsepower Requirements

Results

Three years after installation, Chem-Aqua surveyed a facility that had installed the NCD on a 2,000 ton ammonia refrigeration system with multiple oil chillers. The condenser tubes showed significant scale buildup and the chillers were so badly scaled that oil temperatures were running at 165-175°F instead of at 120-130°F.

Improper "tuning" of the NCD and inadequate filtration were blamed for the failure. The NCD supplier provided a new, properly-tuned device and larger filtration. Over $20,000 was spent to chemically clean the evaporative condensers and chillers so the trial could resume.

Increased Energy Cost and Greenhouse Gas (GHG) Emissions Associated with 1/16" Condenser Deposit			
Refrigeration Load	Increased Energy Cost	GHG Emissions	Equivalent Cars
500 Tons	$26,310	339.8 Tons	58.8
1,000 Tons	$52,620	679.8 Tons	117.6
2,500 Tons	$131,550	1699.5 Tons	294.0

Based on 0.67 kW/ton refrigeration efficiency, electricity cost of $0.06 kW, 1.55 lbs CO_2 emissions per kWh, and 11,560 lbs CO_2 per car per year.

Six years after installation, the condenser tubes were 75% bridged with calcium carbonate scale that was 1/16" to 1/4" thick and the plant was having difficulty keeping the refrigeration system online. They were permitted to remove the NCD and resume chemical treatment. Over $31,000 was spent to chemically clean the systems to restore the head pressure before chemical treatment was implemented. The chemical program has kept the condenser tubes clean and energy-efficient. As of last update, the NCD has been replaced by chemical treatment in all but three facilities within the corporation.

Any environmental benefits gained from use of the NCD were unquestionably offset by increased energy usage and greenhouse gas emissions. Instead of saving money and safeguarding the environment, the NCD resulted in reduced system reliability and higher operating costs.

*Registered trademark of Time Inc.

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